

Lessa?! | لسه؟

Smart Government Queue Tracking Platform

Product Vision Document

Version 1.0 — Hackathon MVP Edition | June 2026

Transforming Government Service Queues Across MENA

1. Product Overview

Lessa?! (لسه؟ — Arabic for "Still?!" or "Not yet?") is a bilingual (Arabic/English) web platform that modernizes the citizen experience at government service branches by enabling online queue slot booking, real-time position tracking, and AI-assisted wait time prediction. Citizens can book their place in line before leaving home, track their position remotely, and receive turn-approaching notifications — eliminating the frustration of physical waiting.

The platform serves five distinct user roles: Guests (public, unauthenticated), Citizens, Counter Staff, Branch Supervisors, and System Administrators. Each role has a purpose-built interface optimized for their specific workflow.

"Lessa?!" echoes the universal Egyptian frustration of asking "Are we there yet?!" — transforming that anxiety into actionable, real-time information.

2. Problem Statement

Government service branches in Egypt and across the MENA region suffer from chronic queue mismanagement. Citizens arrive early, wait hours, and still miss their turn. Branch staff have no real-time visibility into congestion. Supervisors lack the data needed to allocate windows effectively. The result is a systemic erosion of trust in public services.

Core Problems

- Citizens have zero visibility into wait times before arriving
- No mechanism for remote queue booking; presence required to secure a place
- Staff calling systems are often manual, ad-hoc, and error-prone
- Branch supervisors cannot measure window efficiency or identify peak-hour bottlenecks
- No accountability trail for service-level violations or ticket disputes
- Citizens with appointments compete with walk-ins in undefined priority queues
- Language barriers — monolingual systems exclude segments of the population

User Pain Points

Citizen

- "I don't know if it's worth going today — it might be a 3-hour wait."
- "I left work early, arrived at 10am, and still waited until 2pm."
- "I got my number, left to grab coffee, missed my turn."
- "The paper ticket machine was broken — staff were calling names informally."

Counter Staff

- "Sometimes 5 people are waiting at my window but 3 windows next to me are empty."
- "I have to manually cross off numbers and shout queue numbers."
- "When I skip a no-show, there's no record and citizens argue about it."

Branch Supervisor

- "I have no idea how many people we served today, or which window was slowest."
- "I can't prove to management that we need more staff during peak hours."

3. Target Users

User Segment	Primary Interaction	Key Frustration
Egyptian citizens / MENA residents	Book, track, receive alerts	Wasted travel, unpredictable waits
Government counter staff	Call next, mark done/skip	Manual paper-based process, disputes
Branch supervisors	Monitor windows, review metrics	No dashboard, reactive management
System admins	Configure services, manage users	Manual onboarding, no central config
Walk-in guests (no account)	Check congestion, see live times	Unknown wait before deciding to go

4. Value Proposition

Stakeholder	Value Delivered
Citizens	Book remotely, skip physical waiting, get notified before their turn
Counter Staff	Structured digital calling, reduced disputes, clear transaction log
Branch Supervisors	Live monitoring dashboard, configurable windows, end-of-day reports
Government / Ministry	Data-driven capacity planning, accountability trail, modern service image
System Admin	Self-service configuration, RBAC, audit logs

5. Business Goals

1. Reduce average citizen wait time by $\geq 40\%$ within 3 months of deployment
2. Reduce no-show rates through proactive turn-approaching notifications
3. Enable data-driven staffing decisions by providing supervisors with daily/hourly analytics
4. Create a reusable platform deployable across multiple branches/ministries (post-MVP)
5. Establish a public-sector digital transformation reference implementation

6. Success Metrics / KPIs

Target measured 30 days post-launch:

Metric	Target (30-Day Post-Launch)
Daily active bookings per branch	≥ 200
Average wait time reduction	≥ 40% vs. baseline
No-show rate	≤ 15%
Check-in completion rate	≥ 80% of booked tickets
Citizen satisfaction score	≥ 4.0 / 5.0
Notification delivery success	≥ 95%
AI prediction accuracy (±5 min)	≥ 70%
Staff ticket call cycle time	≤ 30 seconds per call

7. Product Principles

Clarity over cleverness

Every citizen interaction must be understandable in under 10 seconds.

Bilingual first

Arabic RTL and English LTR are equally supported, not retrofitted.

Offline-resilient

Degraded mode (no real-time) must still function for basic tracking.

Auditability by default

Every critical action leaves a traceable log entry.

Progressive disclosure

Simple for citizens, powerful for staff and supervisors.

Mobile-first but not mobile-only

Public display boards require desktop rendering.

8. MVP Scope Summary

In Scope

- Citizen registration (phone + name) and login
- Service catalog (defined by admin)
- Queue slot booking and cancellation
- Real-time queue position tracking (SSE or polling)
- Counter staff calling interface
- Branch supervisor monitoring dashboard
- Basic reporting: daily transactions, avg wait, no-show rate

- AI-assisted wait time prediction (heuristic)
- Public display board (current/next tickets, live estimates)
- Bilingual UI (Arabic + English)
- Audit logging for critical actions
- RBAC (5 roles)
- Configurable queue rules (grace period, cancellation cutoff)

Non-Goals (MVP)

- Multi-branch management UI
- Native mobile app (iOS/Android)
- Payment integration for any paid services
- SMS/push notification delivery (in-app only for MVP)
- ML-trained prediction models (heuristic only)
- Integration with external government databases or national ID systems

9. Differentiators vs. Generic Queue Systems

Generic Queue System	Lessa?!
Paper/physical tickets	Digital booking from anywhere
No wait time estimation	AI-assisted predicted wait time
English-only	Full Arabic RTL + English
No citizen notification	Turn-approaching alerts
Basic counter display	Rich public board + supervisor dashboard
No audit trail	Full audit log for every critical action
Single role	5 distinct role-optimized interfaces
No analytics	Daily/weekly/peak-hour reports

10. How the AI Feature Strengthens the Experience

The AI Wait Time Prediction feature transforms Lessa?! from a passive queue tracker into an active decision-support tool. Citizens and guests can see a predicted wait time before deciding to book or travel.

- Reduces unnecessary trips — citizens with a predicted 2-hour wait may choose a less congested time
- Sets expectations accurately — a confident "~18 minutes" beats "number 47 of 60"
- Enables smart notifications — "Your estimated turn is in 12 minutes — head to the branch now"
- Supports supervisor decisions — predicted surge alerts trigger proactive window opening
- Improves over time — even with a heuristic model, historical averages improve accuracy daily

$$\text{MVP formula: Predicted Wait} = (\text{tickets_ahead} \times \text{avg_handling_time_per_service}) / \text{active_windows_count} \times \text{historical_factor}$$

11. Why Bilingual and Public-Sector-Specific Design Matters

Arabic is the primary language of the target user base. Government services typically serve all demographic segments, including elderly citizens and those with limited literacy in Roman-script languages. A monolingual English interface creates real access barriers.

Design Considerations

- Large, clear typography (serving elderly users)
- High-contrast color schemes (accessibility)
- Minimal steps to book (reducing abandonment)
- Trust-signaling UI (official colors, government-appropriate aesthetics)
- RTL-first layout that does not feel like a "flipped" LTR interface

12. Post-MVP Roadmap

Phase	Timeline	Key Features
Phase 2	Month 2–3	Multi-branch, SMS/push notifications, QR code check-in
Phase 3	Month 4–6	ML-based prediction, appointment scheduling, API for 3rd parties
Phase 4	Month 7–12	Ministry-level dashboard, SLA monitoring, citizen rating system
Phase 5	Year 2	Native mobile apps, national ID integration, inter-branch referrals

13. Risks

Risk	Probability	Impact	Mitigation
Real-time feature complexity delays core queue	Medium	High	Use polling fallback if SSE proves complex
AI prediction adds 1 day of dev time	Medium	Medium	Heuristic model keeps this to <2 hours
Arabic RTL layout breaks on some components	Medium	High	Use CSS logical properties; test RTL early
Demo data insufficient to show AI value	Low	High	Pre-seed realistic historical records
Phone OTP service unreliable or costly	High	Medium	Use static test codes for MVP; abstract provider

14. Assumptions

- MVP covers a single branch deployment; the data model is multi-branch ready
- Citizens authenticate via phone number; OTP is simplified or mocked for MVP
- Real-time updates use SSE (Server-Sent Events) for simplicity; WebSocket is Phase 2
- AI prediction is a heuristic model (no ML training required) using queue state + historical averages
- The branch operates during configurable business hours; bookings outside hours are rejected
- Staff are pre-configured by admin; self-registration for staff is Phase 2
- Notifications are in-app for MVP; SMS/push are Phase 2 integrations

15. Constraints

Constraint	Detail
Team	3 engineers/designers
Time	2 days (hackathon delivery)
Infrastructure	Single-server / PaaS deployment (Render, Railway, Heroku)
Budget	Near-zero for external services
Compliance	No PII storage beyond phone number and name (GDPR-lite)
Language	Arabic and English UI required from day one